CHAPTER 12

Role of probiotics in gallstone treatment

Ratnakar Shukla[1], Muskan Chadha[2], Rafia Shekh[3] and Rohit Kumar Tiwari[1]
[1]Department of Clinical Research, Sharda School of Allied Health Sciences, Sharda University, Greater Noida, Uttar Pradesh, India
[2]Department of Nutrition and Dietetics, Sharda University, Greater Noida, Uttar Pradesh, India
[3]Department of Biosciences, Integral University, Lucknow, Uttar Pradesh, India

Introduction

Gallstones (GS) are a common health issue significantly impacting human population globally (Stinton & Shaffer, 2012). In developed countries, 10%−20% of the adult population have been previously diagnosed positively with GS and the prevalence of this disease is constantly increasing (Parra-Landazury et al., 2021/2021; Stinton & Shaffer, 2012). Current evidence indicates that GS is significantly associated with gallbladder cancer (GBC) (Dutta et al., 2019). GS greater than 3 cm in size carry the greatest risk of developing GBC, particularly in symptomatic GS patients (Behari & Kapoor, 2012). Additionally, GS also serves as a major risk factor for various diseases including cholecystitis, empyema, pancreatitis, biliary colic, cholangitis, and bile duct obstruction along with pancreatic and colon cancer (Schernhammer et al., 2003).

Removal of gallbladder (cholecystectomy) is the most common approach used for prevention and treatment of cholelithiasis. However, cholecystectomy is an invasive procedure and may cause serious health concerns like intestinal dysfunctions. Indeed, a previous report has shown that cholecystectomy significantly affects the composition of the gut microbiota (Yoon et al., 2019). Post-cholecystectomy, a number of taxa such as *Prevotella, Desulfovibrio, Barnesiella, Paludibacter, and Alistipes* were found to be decreased and *Bifidobacterium, Anaerostipes*, and *Dorea* were found to be increased as compared to healthy subjects that might result in high levels of secondary bile acids (Wang et al., 2018; Yoon et al., 2019). Other studies have shown that *Blautia obeum*, *Veillonella parvula*, and *Bacteroidetes* were increased after cholecystectomy (Keren et al., 2015;

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Wang et al., 2018; Yoon et al., 2019). It has been noted that alteration in microbial flora after cholecystectomy persists for a long time. Previous studies have shown that cholecystectomy is associated with colon cancer (Frost et al., 2021), gastric cancer (Chen et al., 2014; Fall et al., 2007), and small intestinal cancer (Goldacre et al., 2012). Therefore, future studies must focus on preventive strategies to prevent the formation of GS.

Accumulative evidence has shown that gut microbiota plays an important role in pathogenesis of GS disease (Grigor'eva & Romanova, 2020). The microbial diversity is reduced in patients with GS disease with increased levels of pathogenic bacteria like *Salmonella typhi*, *Helicobacter pylori*, and *Escherichia coli* (Grigor'eva & Romanova, 2020; Keren et al., 2015) which leads to oxidation and epimerization of bile acids further impairing enterohepatic circulations. Thus, these bacteria contribute to the formation of GS and enhance the complications like cholangitis, acute and chronic cholecystitis, and pancreatitis (Fatemi et al., 2018). Major factors leading to formation of GS are malfunctioning of gallbladder motility, metabolism, secretion, and absorption of cholesterol and bile acids (Wu et al., 2013). Probiotics offer a new therapeutic approach to prevent and alleviate the symptoms of GS disease by lowering the serum cholesterol, regulating bile acid metabolism, and restoration of beneficial bacteria and diversity of the microbiota (Grigor'eva & Romanova, 2020).

Gallstones

GS or cholelithiasis is the most common type of gallbladder disease and recent prevalence of GS is increased by more than 10% (Zhang et al., 2022). Approximately 20 million population in the USA (15% of the population) are suffering from GS and frequency of GS is more prevalent in American Indians and Mexican-Americans than in black Americans (Bansal et al., 2014). The prevalence of GS in the European population is 9%–21%, 5%–20% in the Asian population, and 3%–5% in Africans (Acalovschi & Lammert, 2012/2012; Angelico et al., 1997; Bansal et al., 2014; Everhart & Ruhl, 2009). The prevalence of GS in India varies from 2% to 29%. It has been noticed that occurrence of GS is much more frequent in north as compared to other regions of India (Bansal et al., 2014; Dutta et al., 2019). Most of the patients with GS are asymptomatic, while few experience discomfort in the right upper abdomen, dyspepsia, and biliary colic. If GS are not diagnosed or treated in time, the patient may develop acute cholecystitis with infection (Zhang et al., 2022).

The GS is usually constituted by precipitated bile components, which include cholesterol, bilirubin, and bile salts (Njeze, 2013/2013). GS are classified into three major types. The first and the most common among all is the cholesterol stone. The remaining two are black or brown pigmented stones, respectively. It has been found that the cholesterol stones and the black pigmented stones are formed exclusively in the gallbladder, while the brown pigmented stones can be formed anywhere in the biliary tract due to the infection by anaerobic bacteria. Stasis of bile juice is an indispensable factor in the development of GS or bile duct stone (Pak & Lindseth, 2016). GS composition has been reported to vary significantly within different parts of India. In the eastern and northern parts of India, GS have been predominantly found to be either mixed stones or cholesterol stones whereas in southern part of India, pigmented stones are predominant in GS patients (Poddar, 2010).

Studies have previously also shown the association of different types of GS with altered microbiota (Cetta, 1991; Maki, 1966; Swidsinski et al., 1995; Wetter et al., 1994). Maki first reported that bacterial infection may play a key role in pathogenesis of pigmented gallstones (Maki, 1966). This conclusion was drawn owing to the observation that bacterial DNA was present in all brown pigment stones while it was rarely present in black pigment stones or GS with high cholesterol of 70%–95% (Cetta, 1991; Swidsinski et al., 1995; Wetter et al., 1994). Another study also showed that bacteria were cultured from the core of GS in 81% and 77% of cases among patients with gallstone disease (GSD) and GBC, respectively (Hazrah et al., 2004). Both enteric and non-enteric bacteria including *E. coli*, *Salmonella* spp., and *Klebsiella* spp. were cultured from patients with cholelithiasis and only enteric bacteria were cultured from GBC (Hazrah et al., 2004). These observations indicated that chronic infections may be predisposed as a causative factor of GS and subsequently result in the onset of GBC.

Certain factors play a crucial role in the pathogenesis of cholesterol stone formation. These factors include hypersecretion of cholesterol in hepatocytes, increased transition of cholesterol into bile, impaired motility of gallbladder, genetic factors, various intestinal factors affecting the cholesterol absorption, and altered microbiota in the gut (Figueiredo et al., 2017). Wu et al. reported reduced microbial diversity among patients with GSD with decrease in the beneficial genus *Roseburia* and an increase in the Prototype of bacteria with a wide range of pathogenic bacteria like *Salmonella, Vibrio, Helicobacter*, and *Escherichia* (Keren et al., 2015; Wu et al., 2013).

Indeed, increasing evidence has indicated that bacteria play an important role in cholesterol gallstone formation (Wu et al., 2013). Previous *in vivo* studies showed that enterohepatic *H. pylori* elevated the formation of cholesterol GS (Maurer et al., 2005). *In vitro* study also showed that *Helicobacter* spp. can precipitate the calcium salt resulting in GS formation (Belzer et al., 2006). At least one urease-positive *Helicobacte*r either monoinfection or coinfection is necessary to induce the formation of GS (Belzer et al., 2006; Maurer et al., 2005). Interestingly, an 80% prevalence rate of cholesterol GS was reported during *in vivo* studies when mice were infected with *Helicobacter bilis* and coinfected with *Helicobacter hepaticus* and *Helicobacter rodentium*. The prevalence of cholesterol GS formation was reduced to less than 40% in the case of monoinfection with the above-stated bacterium (Maurer et al., 2005). The findings of the study indicated that infection with multiple bacterial species increases the chances of GS formation compared to infection with a single bacterium. These observations press the need for exploring the compositional diversity of biliary microbiota and their effects on GS formation to gain a better understanding of GS pathogenesis (Wu et al., 2013).

Biliary microbiota

The biliary microbiota is a unique and important community of commensal microorganisms that reside within the biliary tract. Biliary microbiota is complex and diverse, comprising a wide range of bacterial species, archaea, fungi, and other microorganisms (Serra et al., 2021). These microbial populations are dynamic and can change in response to environmental conditions and host health status. It also plays an important role in maintaining the homeostasis of the gastrointestinal tract and protecting against the effects of pathogenic microorganisms competing for food and niche (Collins et al., 2023). Studies have shown that infection with fish-borne liver fluke leads to alteration in the composition of biliary microbiota leading to the development of GS along with other diseases such as biliary obstruction, liver cirrhosis, primary sclerosing cholangitis, ischemic colitis, and intestinal metaplasia (Saltykova et al., 2016/2016). Additionally, it has also been reported that the biliary microbiota composition can be affected by diet, lifestyle changes, and various medical interventions (Devkota & Chang, 2015).

Biliary microbiota composition is an essential factor in determining the progression of gallstone disease (Saltykova et al., 2016/2016). In a recent

study, researchers investigated the effects of biliary microbiota composition on biliary cholesterol saturation levels, which play an important role in GS development. The study analyzed intestinal flora from healthy mice and mice with GS. Results showed that mice with GS had higher imbalance of intestinal flora, along with a decrease in biliary cholesterol saturation levels compared to healthy mice (Chen et al., 2021). Thus, dysbiosis of intestinal and biliary microbiota may play a substantial role in GS formation (Chen et al., 2014; Wang et al., 2017). Studies have shown that *H. pylori* and *S. typhi*, the two microbial species commonly found in the gastrointestinal tract, are associated with an increased risk of developing GS disease, which subsequently increases the complication of other GSD, including pancreatitis, cholangitis, and cholecystitis (Grigor'eva & Romanova, 2020; Wang et al., 2018).

The motility of the gallbladder and the expression of mucin genes (MUC1, MUC3, MUC4) can be influenced by pathogenic microbes of the oral cavity through immunomodulatory processes. Furthermore, several pathogens can also serve as one of the factors favoring GS development. Also, the presence of GS can lead to the formation of biofilms and release of pro-inflammatory mediators that have previously been reported to aggravate the inflammatory state (Fremont-Rahl et al., 2013; Grigor'eva & Romanova, 2020). The possible mechanism related to the impact of alterations in biliary microbiota on the development of GS disease is depicted in Fig. 12.1.

A study from 2019 examined the composition and activity of biliary microbiota in patients with and without GS. It was reported that 16S rRNA gene-based sequences from the families *Bacteroidaceae, Prevotellaceae, Porphyromonadaceae*, and *Veillonellaceae* were more abundant among patients with GS, whereas sequences from the family *Propionibacteriaceae* were more prevalent in bile samples within the control group. This report thus established the fact that microbial population is substantially altered in GS patients as compared to healthy individuals (Molinero et al., 2019). Furthermore, previous studies have also substantiated that biliary microbiota usually modulates the expression levels of mucin, host immune response, phospholipases, and oxycholesterols. These observations thus indicated a strong association between biliary microbiota and GS formations, indicating that bacteria may act as an initiating agent causing both cholesterol and pigmented GS (Swidsinski & Lee, 2001). Findings from another study also showed that the patients with GS were significantly associated with higher abundance of gut bacteria like *Proteobacteria* at

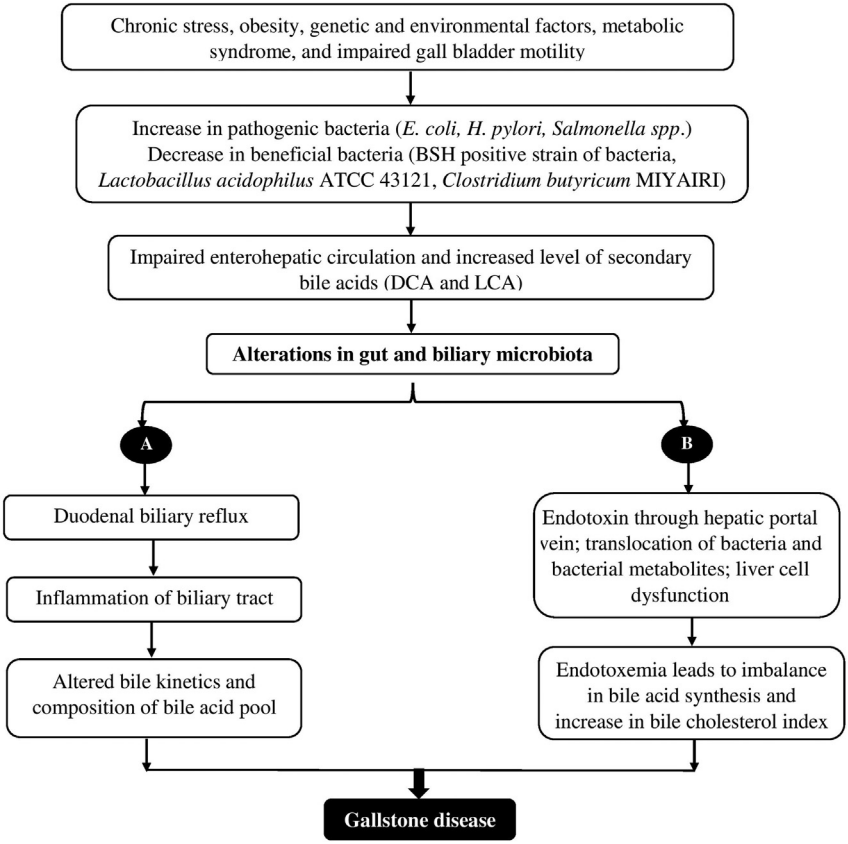

Figure 12.1 ***Impact of alterations in biliary microbiota on GS formation.*** Various extrinsic and intrinsic factors are responsible for different types of cholelithiasis. These factors lead to an increase in number of pathogenic bacteria and decrease in beneficial bacteria. Pathogenic bacteria disrupt the enterohepatic circulation and impair the metabolism of lipids and cholesterol, leading to increased levels of secondary bile acids like deoxycholic acids (DCA) and lithocholic acids (LCA). Impaired enterohepatic circulations result in alterations in gut and biliary microbiota. (A) Translocation of bacteria and bacterial-derived metabolites from the gut due to increased permeability of intestinal epithelial cells leads to portal endotoxemia, followed by hepatic dysfunction. Endotoxemia leads to an imbalance in bile acid synthesis, and an increase in bile cholesterol index results in the formation of GS. (B) The duodenal–biliary reflux pathway facilitates bacterial entry into the bile duct from the duodenum and contributes to bacterial invasion leading to inflammation of biliary tract. This, in turn, promotes the changes in biliary kinetics and composition of bile acid pool, followed by alterations in biliary microbiota resulting in gallstone disease.

phylum level followed by a reduced population of *Faecalibacterium, Lachnospira*, and *Roseburia* at genus level. Approximately 70% of operational taxonomic units of gut microbiota among patients with GS were present in biliary microbiota, and diversity of biliary microbiota was significantly higher than in the gut. This further suggests that biliary microbiota imparts a significant influence on GS formation. Furthermore, analysis of the biliary microbiota in GS patients revealed that almost 33.9% of constituents of biliary microbiota were linked to recognized bacterial species such as *Propionibacterium acnes, Bacteroides vulgates*, and *Pseudomonas putida*, etc., and played an important role in GS development (Wu et al., 2013).

In comparison to individuals with pigmented GS, patients with cholesterol GS have a decreased frequency of bacterial isolates in bile (Stewart et al., 2005). The percentage of bacterial isolates cultivated from the bile in brown cholesterol, pigmented, and black pigmented GS patients ranged from 53%–83%, 9%–34%, and 9%–19%, respectively (Abeysuriya et al., 2008; Stewart et al., 2005). The oxidation and epimerization of bile acids by intestinal microbes such as *Clostridium, Bifidobacterium, Peptostreptococcus, Bacteroides, Eubacterium*, and *E. coli* can also plausibly impair enterohepatic circulation and cause GS through immune modulation which subsequently affects the gallbladder motility and expression of mucin genes including MUC1, MUC3, MUC4 (Grigor'eva and Romanova, 2020). Studies have revealed an increased population of Gram-negative anaerobic and facultative bacteria in gallbladder bile may contribute to the GS formation and subsequently increase the risk of biliary tract cancer (Swidsinski & Lee, 2001). It has also been previously reported that, as compared to controls, GS patients have a higher average of fecal bile acid levels with significantly reduced microbial diversity. In fecal isolates from GS patients, it was found that there was a significant decline in the helpful genera, such as *Roseburia*, with a concomitant increase of the undeveloped species *Oscillospira*. In light of this and other evidence from this study, it was inferred that these two genera may act as indicator for the development of symptomatic GS (Keren et al., 2015).

Current studies have shown that *Clostridium, Bifidobacterium*, and *Enterorhabdus* are involved in the bile acids metabolism, and excess bile acids in the intestine disrupt enterohepatic circulation and may cause GS formation (Grigor'eva and Romanova, 2020). Higher level of secondary bile acid, including deoxycholic acid (DCA) and lithocholic acid (LCA), was previously found to be associated with increased risk of GSD (Berr et al., 1996). Chen et al. reported that a higher level of DCA was found to be significantly higher in mice with GS compared to the control (Chen

et al., 2021). Previous results have also shown that DCA acts as a Farnesoid X receptor (FXR) agonist increases cholesterol saturation in gallbladder and bile acid hydrophobicity that leads to decrease in the solubility of cholesterol in bile and increases the risk for GS formation (Heaton, 2000). *Clostridium* spp. has 7α-dehydroxylation enzymatic activity, which can convert primary bile acids into secondary bile acids such as DCA (Berr et al., 1996). Chen et al. showed that abundance of *Clostridium* and *Bifidobacterium* were significantly higher in GS mice than in control mice (Chen et al., 2021). The level of 7α-dehydroxylating bacteria was 42 times higher in patients with GSD than those without it (Ridlon et al., 2010). These exhaustive results have substantiated the role of gastrointestinal and biliary microbiota in GS formation. Nevertheless, these evidences have further potentiated the need for further explorations to evaluate the efficacy of gastrointestinal and biliary microbiota analysis as a predictive marker for gastroenterological disorders.

Probiotics

Probiotics are live microorganisms that when consumed in adequate amounts, confer a health benefit to the host. They have been used since ancient times, but in recent years, they have become increasingly popular (Gratz et al., 2010). The human gastrointestinal tract is home to a large number of bacterial species that are critical for health and well-being. Probiotics are an essential part of gut health, and evidences have indicated that they promote a healthy digestive system and boost the natural defenses against illness (Gratz et al., 2010). Probiotic bacteria enhance breakdown of ingested food, nutrient absorption, and vitamin production, including vitamins B and K (Gu & Li, 2016). Intriguingly, reports have further suggested that probiotics can help to reduce symptoms of gallstone, diarrhea, constipation, irritable bowel syndrome, and other gastrointestinal issues (Amara & Shibl, 2015; Han et al., 2022). Probiotics for gallstone treatment represent an emerging field in nutrition and alternative medicine. GS are hard deposits in the gallbladder that cause pain with several other clinical presentations (Lammert et al., 2016/2016). The use of probiotics to treat GS is based on the concept of "friendly bacteria" that help to protect against pathogens *H. pylori, S. typhi*, and *E. coli* , which help in promoting the useful gastrointestinal and biliary microbiota that would subsequently lower the population of GS-promoting microbes (Binda et al., 2022; Rajkumar et al., 2015).

Recently, a study showed the preventive effect of *Lactobacilli* (*L. acidophilus* ATCC 43121 and *L. fermentum* MF27) on GS formation in lithogenic diet (LD)-induced C57Bl/6J mice models. After four weeks of therapy, LD with *L. acidophilus* ATCC 43121 (LD + P1) and LD with *L. fermentum* MF27 (LD + P2) reduced the cholesterol level, which was found to be associated with a decreased level of 3-hydroxy-3-methylglutary coenzyme A reductase in the liver compared to LD group. Due to these traits, the expression of mucin-forming genes, including MUC5AB and MUC5B, was also found to be reduced in the LD + P1 and LD + P2 groups than LD group (Oh et al., 2021). Thus, it may be inferred that impeding mucin overproduction can be a preventive approach for GS formation in gallbladder (Chuang et al., 2012). Therefore, both *Lactobacilli* spp. and its regular therapeutic usage can be beneficial in avoiding gallstone formation as they both function to limit the production of cholesterol GS (Oh et al., 2021).

Previous meta-analysis report has indicated that the rate of GS formation reached up to 30% after bariatric surgery (Magouliotis et al., 2017; Uy et al., 2008). Recently a prospective randomized controlled trial study reported by Han et al. evaluated the effect of probiotic (1 gm *Clostridium butyricum* MIYAIRI accounting for approximately 5×10^9 colony-forming units), ursodeoxycholic acid (UDCA, 300 mg) and digestive enzyme group (Biotase 3 tablets-Biodiastase 30 mg + lipase 65 mg + newlase 10 mg) in prevention of GS among patients with bariatric surgery. UCDA is more frequently used in the prevention of cholesterol stones in patients with GS (Castro-Torres et al., 2015; Han et al., 2022). After six months, the frequency of GS formation in probiotic group was found to be 15.2%, 17.6% in the UDCA group, and 29.1% in digestive enzyme group. Although the results showed insignificant differences in GS formation; the study suggested that the efficacy of probiotics is not inferior to UDCA and can thus be considered an alternative treatment for prophylaxis of GS formation in post-bariatric patients (Han et al., 2022). *C. butyricum* MIYAIRI is considered an important probiotic that has been utilized for reducing plasma cholesterol levels in cholesterol-fed rats by up to 33% (Ariyoshi et al., 2020; Han et al., 2022). Intriguingly, previous report has also shown that oral administration of *C. butyricum* MIYAIRI decreased the GS index, incidence, and cholesterol content of GS in cholelithiasis mice models by 46%, 43%, and 46%, respectively, compared to the untreated control group (Takeda et al., 1983). Monoinfection with *C. butyricum* MIYAIRI no. 588 strain has been previously reported to

increase the excretion of bile acid and inhibit GS formation in JCL:ICR male germ-free mice. Therefore, it may be concluded that the composition, secretion, and metabolism of biliary lipid content is affected significantly by intestinal microflora which subsequently can ameliorate GS formation in *in vivo* model (Hosomi et al., 1982).

Certain intestinal microbial flora produce cholesterol reductase which helps in catalyzing the conversion of cholesterol to insoluble coprostanol (Horáčková et al., 2018) which is a saturated form of cholesterol (Juste & Gérard, 2021) and it accounts for almost 50% of total sterol excreted in feces and thus reduces exogenous level of cholesterol (Horáčková et al., 2018). A meta-analysis study showed that probiotics supplements (VSL #3, *L. acidophilus, Bifidobacterium animalis subsp. lactis*, and *Lactiplantibacillus plantarum* group) may significantly lower the total serum cholesterol level (Wang et al., 2018). Another study showed that consumption of bile salt hydrolase (BSH)-positive strain of *Lactobacillus* significantly reduced the cholesterol in patients with hypercholesterolemia (Jones et al., 2012). *Lactococcus* is one of the most common probiotics that is used in human gastrointestinal tract. Members of the *Lactococcus* were found to increase in patients with GS. In contrast to the BSH activity of *Lactobacillus* strains, the underlying physical activity of *Lactococcus* remains unknown and needs to be explored (De Smet et al., 1995; Wu et al., 2013).

The effect of probiotics can be mediated by by metabolites such as short-chain fatty acids that have been shown to exert antiinflammatory effects, suggesting that these may be useful for reducing inflammation and helping in fewer episodes of discomfort associated with GS (Cristofori et al., 2021). Probiotics may work to dissolve GS by regulating the levels of bile and cholesterol via binding to the Farnesoid X receptor (bile acid receptor). *Bifidobacterium breve* and *Lactobacilli* are two probiotics that can reduce B-glucuronidase activity. *Lactobacilli* can also suppress urease activity. In combination, these probiotics can be taken in conjunction to manage the variables that lead to gallstone formation and may avoid cholecystectomy (Kesharwani et al., 2022).

The bile salt hydrolase activity of microorganisms may promote the hydrolysis of bile acids into simpler compounds and deconjugation of bile acids into free bile acids (Grigor'eva and Romanova, 2020). For instance, 7α-dehydroxylation of cholic acid and chenodeoxycholic acid is mediated by the BSH activity of microorganisms to form deoxycholic acid and lithocholic acid, respectively, which makes bile salts

available as substrates for further modifications by gut microbiota (Cai & Chen, 2014; Grigor'eva & Romanova, 2020). Moreover, BSH activity enables the microorganism to survive in biliary environment by contributing resistance towards the bile acid at low pH (Kawai et al., 2002). Therefore, BSH activity must be explored further for its plausible probiotic efficiency. BSH activity has also been previously observed in 273 strains of *Bifidobacterium* and *Lactobacillus* and in *Clostridium* spp., *Bacteroides* spp., *Enterococcus* (Ridlon et al., 2016) but was absent in *L. lactis*, *Leuconostoc mesenteroides,* and *Streptococcus thermophilus* (Masco et al., 2007). Nevertheless, the effects of the biliary microenvironment are considered to be hostile for most of the bacteria, however, on bile bacteria, its effect remains yet to be explored (Grigor'eva & Romanova, 2020).

Results from previous studies have shown that high variability in resistance can be found within a genus or species with respect to biliary environment like bile acid, bile salt, and low pH (Begley et al., 2005). Chateau et al. studied the effect of bile salts on a collection of 38 strains of *Lactobacillus*, particularly *L. rhamnosus*, isolated from a probiotic bacterial consortium and showed the variability of resistance. Among 22 *L. rhamnosus* strains, three strains were classified as resistant, five as tolerant, three as low tolerant, and eleven as sensitive (Chateau et al., 1994). Ibrahim and Bezkorovainy further studied the survival of *Bifidobacterium* strains in bile and found that *B. infantis* showed optimum survival rates, whereas *B. longum* exhibited the lowest survival rates (Ibrahim & Bezkorovainy, 1993). Contrastingly, Clark and Martin demonstrated that *B. longum* was much more tolerant than *B. infantis* in bile (Clark & Martin, 1994/1994). These results indicated that the type of strain-specific trait with respect to bile tolerance must be considered for the selection of probiotics, and tolerance of species cannot be generalized (Begley et al., 2005).

Mucin is well known to maintain the integrity of intestinal epithelial cells and protect the intestinal mucosal barrier (Grondin et al., 2020). An abnormal level of mucin has also been reported in several tumors (Paone & Cani, 2020). Previous results showed that mucin-4 is involved in GS formation. Mucin aids GS formation by acting as a pro-nucleating agent. The hydrophobic binding sites present within the core of this polypeptide harbor a favorable environment for cholesterol monohydrate nucleation from hyper-saturated bile (Bar Dayan et al., 2004). Glucosamine is a precursor in the synthesis of mucin (Di Ciaula et al., 2018; Hu et al., 2021). Chen et al. showed that a higher level of glucosamine was found to be

significantly higher in mice with GS than in control. Bacterial genera *Intestinimonas, Ruminiclostridium, Oscillibacter, Bifidobacterium, Tyzzerella*, GCA-900066225, *Ruminococcaceae*_UCG-009, [*Ruminococcus*] _*torques*_group, *Coprococcus*_2, *Harryflintia*, and *Soleaferrea idatus* were shown to have a positive correlation with glucosamine levels during this study (Chen et al., 2021). These bacterial genera were found to be significantly higher in the gut of mice with GS. Interestingly, two genera (*Prevotellaceae*_UCG-001 and *Prevotellaceae*_Ga6A1_group) were found in reduced levels within mice with GS compared to the control and had a negative correlation with glucosamine (Chen et al., 2021). This observation indicated that bacteria belonging to *Prevotellaceae*_UCG-001 and *Prevotellaceae*_Ga6A1_group plausibly may have preventive roles in GSD. Thus, *Prevotellaceae* might be considered as a potential probiotic as it is involved in the synthesis of short-chain fatty acids (SCFAs) (Xie et al., 2020; Zhu et al., 2020). It has been reported that SCFAs protect the intestinal epithelial cells from intestinal pathogens and strengthen gut barrier functions (Venegas et al., 2019). Previous studies have shown that *Prevotellaceae* might be helpful in the prevention of GS formation via regulating bile salt composition and protecting the gut barrier function (Chen et al., 2021; Guo et al., 2021). However, *Prevotellaceae* can be used as a noninvasive observation index for the detection of litholytic drugs remains to be further validated (Chen et al., 2021).

Evidence of previous studies has shown that there is an association of GSD with diet and biliary microbiota. Biliary microbiota is reported to be significantly altered in GS patients (Gutiérrez-Díaz et al., 2018). Consumption of dairy food was found to negatively associated the population of *Bacteroidaceae* and *Bacteroides*, and several types of fibers, phenolic compounds, and fatty acids were found to be associated with abundance of *Bacteroidaceae, Chitinophagaceae, Propionibacteraceae, Bacteroides*, and *Escherichia, Shigella* indicating a strong linkage in between dietary consumption and GS formation (Gutiérrez-Díaz et al., 2018). Thus, a healthy diet and specific nutritional interventions, including probiotics and prebiotics with concomitantly increased fiber intake may lead to restoration of beneficial bacteria by elevating the diversity of gastrointestinal and biliary microbiota and thus preventing GS formation (Rajkumar et al., 2015; Requena et al., 2018). The function of probiotics in prevention and inhibiting the GS formation is depicted in Fig. 12.2.

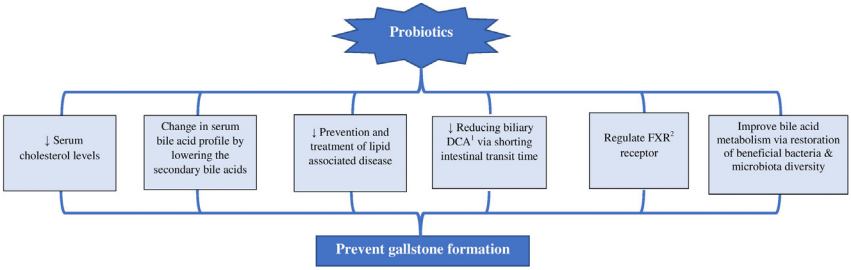

Figure 12.2 ***Role of probiotics in prevention of GS formation.*** [1]*DCA*, Deoxycholic acid; [2]*FXR*, Farnesoid X receptor.

Conclusions

Studies have supported the notion that supplementing the diet with probiotics can reduce the size of GS, prevent their formation, and can even improve the overall health of the gallbladder. Manipulation of altered microbiota with probiotics promotes rapid compensation of the healthy gut microbiota. Both probiotics and prebiotics inhibit the growth of potentially pathogenic bacteria, including *H. pylori, S. typhi*, and *E. coli* thereby promoting the growth of beneficial bacteria such as *Lactobacillus sp., Bifidobacterium sp*., etc. Additionally, probiotics also regulate metabolic functioning by improving the bile acid, glucose metabolism and maintaining homeostatic cholesterol levels. These attributes ultimately reduce the risk of GS formation. Findings from the current studies have evidently suggested that probiotics confer a new non-pharmacological alternative for the prevention and treatment of lipid-associated disease, including **hyperlipidemia and gallstones**. Data on the role of probiotics in the prevention and treatment of GS are very limited. Therefore, both preclinical and clinical studies need to be carried out further to gain a better and in-depth understanding of the positive effects of probiotics in the therapeutic management of GS.

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